Tanish

Roll No.: 123108004
B.Tech, Artificial Intelligence and Machine Learning
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Education

•National Institute of Technology, Kurukshetra — *Haryana*
B.Tech, Artificial Intelligence and Machine Learning — CGPA: 7.2947 (Till 5th Semester)

•MLS DAV Public School — *Narnaul, Haryana (2023)*
12^{th} Grade — Percentage: 87%

•MLS DAV Public School — *Narnaul, Haryana (2021)*
10^{th} Grade — Percentage: 96%

Personal Projects

•3D Lung Tumor Detection using Deep Learning and Generative AI — *Feb 2026 – Present*
Designed a 3D medical imaging pipeline for automated lung tumor detection from CT scans.

- Processed volumetric CT scan data into structured 3D segmented slices for supervised learning.
- Implemented Generative AI-based data augmentation to address class imbalance and improve model generalization.
- Trained convolutional deep learning models for tumor detection on augmented 3D medical imaging dataset.
- Evaluated model performance using Accuracy, Precision, Recall, F1-Score, and confusion matrix analysis.

•Multimodal DDoS Attack Detection using ML and Deep Learning — *Jan 2026 – Present*
Developed a hybrid Intrusion Detection System (IDS) for real-time DDoS attack classification.

- Processed CICIDS2017 and CICDDoS2019 datasets (80+ flow features) for binary and multi-class attack detection.
- Engineered statistical and temporal traffic features including flow duration, packet/byte rate, and TCP flag metrics.
- Designed multimodal architecture combining Random Forest and XGBoost with LSTM for enhanced pattern learning.
- Applied feature scaling, dimensionality reduction (PCA), and class imbalance handling (SMOTE) to improve generalization.
- Reduced False Positive Rate (FPR) via threshold optimization while maintaining high Recall and ROC-AUC.
- Built end-to-end pipeline in Python using Scikit-learn, TensorFlow, Pandas, and NumPy.

•Automatic License Plate Recognition using YOLO and OCR — *Nov 2025*
Developed an end-to-end deep learning pipeline for real-time license plate detection and text extraction.

- Built a two-stage system using YOLO (Ultralytics) for license plate detection and PaddleOCR for text recognition.
- Implemented detection → plate cropping → OCR pipeline with preprocessing (grayscale conversion, image upscaling).
- Applied transfer learning, layer freezing, and fine-tuning to improve mAP and convergence speed.
- Trained on custom YOLO-format dataset in Google Colab with automated data.yaml configuration.
- Evaluated performance using mAP@50, Precision, and Recall; exported inference-ready model weights.

•Fuzzy Water Quality Checker — *April 2025*
Developed a fuzzy logic-based system to compute WQI from environmental parameters.

- Built using Python and Streamlit for interactive web UI.
- Designed fuzzy membership functions for pH, DO, BOD, Nitrate, and Temperature.
- Applied fuzzy inference rules to classify water into Poor, Moderate, and Good.
- Deployed app on Streamlit Cloud using GitHub integration.

Technical Skills

Core CS: Operating Systems, Object Oriented Programming, DBMS, Distributed Computing
Languages: Python, C/C++, SQL (MySQL), JavaScript, HTML/CSS
Developer Tools: Git, VS Code, Visual Studio, Jupyter Lab, Google Colab
Libraries: Pandas, NumPy, Matplotlib, Scikit-learn (sklearn), TensorFlow, Keras
Machine Learning: Supervised Learning, Deep Learning (RNN, LSTM, CNN), Generative AI , Feature Engineering, Model Evaluation
Computer Vision: Image Preprocessing, YOLO , PaddleOCR, Segmentation, Object Detection

ACHIEVEMENTS

- **Solved 200+ problems** of Data Structures and Algorithm across Leetcode
- Secured an All India Rank under 15,000 in JEE Main Examination
- Winner of District-level Quiz Competition for 2 consecutive years and achieved 2^{nd} position at State level
- Secured All India Rank 34 in ANTHE (Aakash National Talent Hunt Exam)